

A Counterexample to the Converse Between Two p -Universal Representation Notions

arXiv automatic math run `fa_banach_001`

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Status

This packet gives a candidate counterexample to the first open remark after Definition 3.7 in Neufang–Runde [1]. It is a counterexample to a converse between two definitions of p -universal representation. It does not address the later open questions in the source paper about Herz–Schur multipliers or complete isometries.

Source Question

The source paper defines a representation $(\pi_u, E_u) \in \text{Rep}_p(G)$ to be p -universal if it contains every small representation in $\text{Rep}_p(G)$. Immediately afterwards, the authors write:

the desired properties.

□

Definition 3.7. Let G be a locally compact group, and let $p \in (1, \infty)$. We say that $(\pi_u, E_u) \in \text{Rep}_p(G)$ *p -universal* if it contains every small representation in $\text{Rep}_p(G)$. We write $\text{UPF}_p(G)$ instead of $\text{PF}_{p, \pi_u}(G)$ and call the elements of $\text{UPF}_p(G)$ *universal p -pseudofunctions*.

Remarks. 1. Since cyclic representations in the sense of [Run 2] are small, a p -universal representation according to Definition 3.7 is also p -universal in the sense of [Run 2, Definition 4.5]. We do not know if the converse is also true.

2. There are indeed n -universal representations: this can be seen as in the example

Thus the extracted question is whether the older p -universal notion from Runde [2, Definition 4.5], where one contains every cyclic representation, implies the stronger small-representation universality of Definition 3.7 in [1].

Definitions Used

For a locally compact group G , $\text{Rep}_p(G)$ denotes the representations of G by invertible isometries on QSL_p -spaces. A representation (ρ, F) is contained in (π, E) if it is isometrically equivalent to a subrepresentation.

In Runde’s older sense, (π, E) is p -universal if it contains every cyclic representation in $\text{Rep}_p(G)$, where cyclic means that some vector ξ has dense cyclic span $\overline{\pi(L_1(G))\xi} = E$ [2, Definition 4.5].

In Neufang–Runde’s stronger sense, (π, E) is p -universal if it contains every small representation, where small means cardinality at most $\text{card}(L_1(G))^{\aleph_0}$ [1, Definition 3.7].

Proof Intuition

For the trivial group $G = \{e\}$, the algebra $L_1(G)$ is just $\mathbb{C}$, and its action in any representation is scalar multiplication. Therefore a cyclic representation can only be one-dimensional. The one-dimensional trivial representation consequently contains all cyclic representations. But the stronger 2007 definition asks for every small representation, and the trivial representation on ℓ_p^2 is small. A one-dimensional space cannot contain it.

Theorem 1. *For every $p \in (1, \infty)$, the converse in the first remark after Definition 3.7 of [1] is false. More precisely, there is a representation that is p -universal in the sense of [2, Definition 4.5] but is not p -universal in the sense of [1, Definition 3.7].*

Proof. Let $G = \{e\}$ be the trivial locally compact group. Then $L_1(G) \cong \mathbb{C}$. Every representation of G on a Banach space is the trivial representation, and the integrated $L_1(G)$ -action is scalar multiplication:

$$\pi(a)\xi = a\xi \quad (a \in \mathbb{C}, \xi \in E).$$

Let $(\pi, E) \in \text{Rep}_p(G)$ be cyclic in Runde’s sense. Then there is $\xi \in E$ such that

$$E = \overline{\pi(L_1(G))\xi} = \overline{\mathbb{C}\xi}.$$

Hence E is either zero-dimensional or one-dimensional. Every nonzero one-dimensional complex Banach space is isometrically isomorphic to $\mathbb{C}$, and the group action is necessarily trivial. Therefore the one-dimensional trivial representation

$$(\mathbf{1}, \mathbb{C}) \in \text{Rep}_p(G)$$

contains every cyclic representation in $\text{Rep}_p(G)$. It is p -universal in the sense of [2, Definition 4.5].

We now check that it is not p -universal in the stronger sense of [1, Definition 3.7]. Let $F = \ell_p^2$, realized as $L_p(\{1, 2\})$ with counting measure, and let ρ be the trivial representation of G on F . Then F is a QSL_p -space, so $(\rho, F) \in \text{Rep}_p(G)$. It is small: $\text{card}(L_1(G)) = \text{card}(\mathbb{C})$, and

$$\text{card}(F) = \text{card}(\mathbb{C}) \leq \text{card}(\mathbb{C})^{\aleph_0} = \text{card}(L_1(G))^{\aleph_0}.$$

Thus (ρ, F) is one of the small representations that must be contained by any p -universal representation in the sense of Definition 3.7.

But (ρ, ℓ_p^2) is not contained in $(\mathbf{1}, \mathbb{C})$, since containment would require an isometric equivalence of ℓ_p^2 with a closed subspace of the one-dimensional space $\mathbb{C}$, which is impossible by dimension. Hence $(\mathbf{1}, \mathbb{C})$ is old cyclic-universal but not small-universal. $\square$

Verification Notes

The example uses only the trivial group, so no nontrivial continuity or integrability issue is hidden. The representation on ℓ_p^2 is a representation on an actual L_p -space, hence certainly on a QSL_p space. The smallness estimate is exactly the cardinal bound in the source definition. The failure of containment is purely dimensional.

Novelty Check

Local run indexes were searched for 0711.2057, the paper title, QSL_p , p -operator-space, and row/column keywords; no existing packet or attempt for this source question was found. Web searches on June 19, 2026 for phrases including “Column and row operator spaces over QSL_p -spaces we do not know”, “universal p -pseudofunctions p -universal cyclic small representations”, and related Herz–Schur/ $A_p(G)$ terms did not locate a published answer to this converse. The result should be reviewed as a simple new counterexample to the definitional converse, not as progress on the paper’s deeper multiplier questions.

Human Review Recommendation

Check that Runde’s Definition 4.5 indeed allows the one-dimensional trivial representation to count as p -universal for $G = \{e\}$, and that the zero representation, if included, causes no exception. The main conclusion then follows immediately.

References

- [1] M. Neufang and V. Runde, *Column and row operator spaces over QSL_p -spaces and their use in abstract harmonic analysis*, arXiv:0711.2057.
- [2] V. Runde, *Representations of locally compact groups on QSL_p -spaces and a p -analog of the Fourier–Stieltjes algebra*, arXiv:math/0402018.